CS 210: Worksheet 15 *Dr. Mudassir Shabbir, October 15, 2026*

Student ID 1: _____ **Seat Location 1:** row:_____ col:_____

Student ID 2: _____ **Seat Location 2:** row:_____ col:_____

Closures of relations. Write $r(R)$, $s(R)$, and $t(R)$ for the reflexive, symmetric, and transitive closures on the stated set.

1. *Easy*

Let $A = \{1, 2, 3, 4\}$ and $R = \{(1, 1), (1, 2), (2, 3), (3, 3)\}$.
Find $r(R) \setminus R$, then write $r(R)$.

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2. *Easy*

On $A = \{a, b, c\}$, let $R = \{(a, a), (a, b), (b, a), (b, c)\}$.
Find R^{-1} and $s(R) \setminus R$.

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3. *Moderate*

On $A = \{a, b, c, d\}$, let $R = \{(a, b), (b, c), (c, a)\}$.
Find $t(R)$. How many pairs does it contain?

4. *Moderate*

Let $R \subseteq A \times A$ and $S = R \cup R^{-1}$.

(a) Prove that S is symmetric and contains R.

(b) Let T be any symmetric relation with $R \subseteq T$. Prove $S \subseteq T$. Why does this establish $S = s(R)$?

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5. *Challenge*

On $A = \{1, 2, 3\}$, let $R = \{(1, 2), (2, 3)\}$.

(a) Compute $s(t(R))$.

(b) Compute $t(s(R))$.